

CHICAGO ROCKFORD, IL, USA

WMO: 725430

Lat: 42.193N

Lon: 89.093W

Elev: 223

StdP: 98.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-18.0	-25.7	0.4	-20.4	-22.9	0.5	-17.6	11.6	-2.2	10.6	-5.1	3.2	290	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.6	23.5	31.1	22.8	29.7	22.1	25.5	30.5	24.4	28.9	23.4	27.9	5.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.4	28.5	22.9	18.1	27.6	21.9	17.0	26.1	79.2	30.5	74.7	29.0	70.6	28.0	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.3	8.4	-25.3	34.9	4.0	1.8	-28.1	36.2	-30.5	37.3	-32.7	38.3	-35.6	39.5		
(5)			-25.6	27.1	3.9	1.1	-28.3	27.9	-30.6	28.6	-32.8	29.2	-35.6	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-5.9	-3.8	2.8	9.5	15.6	21.1	23.0	22.0	18.1	11.2	3.7	-2.7
	DBStd	11.36	6.87	6.61	6.25	5.10	4.71	3.68	3.10	2.85	4.28	5.04	5.43	6.20
	HDD10.0	1833	494	386	237	68	7	0	0	0	1	46	198	396
	HDD18.3	3628	752	618	483	268	113	16	2	4	55	227	438	653
	CDD10.0	1698	0	1	14	53	179	332	404	373	246	84	10	1
	CDD18.3	452	0	0	1	3	28	97	148	119	49	7	0	0
	CDH23.3	4106	0	0	8	48	291	911	1392	949	440	68	0	0
	CDH26.7	1322	0	0	1	8	82	311	499	280	127	14	0	0
Wind	WSAvg	4.0	4.3	4.4	4.6	4.9	4.2	3.5	3.1	2.9	3.2	3.9	4.3	4.1
Precipitation	PrecAvg	929	37	32	64	98	99	121	97	100	103	70	66	50
	PrecMax	1435	118	78	143	252	208	301	226	344	271	211	140	128
	PrecMin	591	5	1	16	25	12	12	19	12	1	3	10	9
	PrecStd	186	27	20	31	49	44	65	51	71	74	51	35	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	15.6	24.4	27.8	31.6	34.0	35.6	33.5	32.3	28.6	20.9	15.2
		MCWB	9.8	11.4	17.1	17.2	20.4	23.2	25.4	24.1	22.7	20.1	14.5	13.7
	2%	DB	7.4	10.3	19.0	24.5	29.0	31.9	32.7	31.4	30.1	25.1	17.1	11.1
		MCWB	5.4	7.2	13.7	15.8	19.7	22.9	24.5	23.1	21.9	17.5	12.7	9.2
	5%	DB	4.6	7.0	15.5	21.4	26.7	30.2	31.0	29.6	27.9	22.2	14.5	7.7
		MCWB	3.1	4.6	11.2	13.7	18.6	21.9	23.5	22.6	20.4	16.3	11.4	5.9
	10%	DB	2.6	4.4	12.0	18.7	24.3	28.3	29.5	28.0	26.0	19.6	12.1	5.2
		MCWB	1.4	2.6	8.1	12.3	17.4	20.9	22.5	21.5	19.2	15.0	8.9	3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	12.2	17.6	18.8	22.8	25.6	27.4	26.0	24.7	21.4	16.0	13.9
		MCDB	11.4	14.1	23.6	24.6	28.6	30.9	32.2	31.2	29.9	26.3	18.8	15.0
	2%	WB	5.6	7.7	14.8	16.7	21.2	24.1	26.0	24.8	23.0	19.2	13.9	9.3
		MCDB	7.2	10.0	18.4	22.5	26.5	29.4	30.8	29.3	27.4	22.8	16.2	10.5
	5%	WB	3.2	4.8	11.8	15.0	19.9	23.1	24.9	23.8	21.7	17.5	11.6	6.2
		MCDB	4.5	6.8	14.8	20.3	24.8	28.6	29.6	27.7	26.2	21.1	14.0	7.4
	10%	WB	1.4	2.8	8.3	12.9	18.5	22.0	23.7	22.7	20.5	15.5	9.2	3.6
		MCDB	2.6	4.3	11.7	17.8	23.0	26.7	28.0	26.5	24.4	19.1	11.8	5.1
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.5	9.8	11.4	11.5	11.1	10.8	10.6	11.8	10.9	9.0	7.5
		MCDBR	9.7	11.0	13.9	15.8	14.3	12.9	12.0	11.9	13.5	14.2	12.6	10.0
		MCWBR	8.2	8.4	9.2	9.1	7.1	6.0	5.6	5.6	6.4	7.9	9.0	8.3
	5% WB	MCDBR	9.1	10.4	12.8	13.9	12.0	11.3	10.9	10.2	11.5	11.9	11.5	9.1
MCWBR		8.0	8.4	9.3	9.1	6.9	6.1	6.1	5.8	6.2	8.2	9.2	8.2	
Clear-Sky Solar Irradiance	taub	0.285	0.304	0.334	0.386	0.414	0.428	0.440	0.419	0.391	0.341	0.301	0.282	
	taud	2.397	2.359	2.396	2.263	2.237	2.244	2.202	2.258	2.313	2.454	2.505	2.439	
	Ebn at Noon	863	900	912	882	862	849	833	839	838	848	843	836	
	Edh at Noon	79	96	105	130	137	136	141	129	114	87	71	70	
All-Sky Solar Radiation	RadAvg	1.70	2.67	3.68	4.64	5.39	6.11	6.09	5.36	4.45	2.90	1.88	1.38	
	RadStd	0.21	0.32	0.34	0.38	0.38	0.45	0.34	0.33	0.37	0.35	0.16	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+48
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page